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BAGONG PILIPINAS

CC 101 – Introduction to Computing

"Know Your Network Settings": Exploring Windows 11 Network Configuration Tools

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Computer Unit Used (Desktop/Laptop, brand): Laptop, ACER	**Date: 10/19/2026**

I. Objectives

At the end of this activity, you should be able to:

- Locate network adapter information using Windows 11 Settings, the Control Panel, Command Prompt (CMD), and PowerShell.
- Use **ipconfig**, **getmac**, **hostname**, **ping**, **netstat**, **route print**, and **systeminfo** in CMD, and **Get-NetAdapter**, **Get-NetIPAddress**, and **Test-Connection** in PowerShell.
- Interpret a sample **ipconfig /all** output from a networked computer: identify its IP address, subnet mask, default gateway, and DNS servers, and classify the address as public or private.
- Relate each tool and finding to the layer of the OSI model where it works.
- Design a simple network and IP address plan for the computer laboratory.

II. Materials Needed

- A computer running Windows 11 (a network or Internet connection is **not** required for this activity)
- This worksheet (printed copy, or the digital file to fill out and submit)
- Week 6 lecture notes, for reference on IP addresses, protocols, the OSI model, and network devices

III. Background

A computer does not need to be connected to a network to have network settings. Every PC has one or more **network adapters** (NICs), each with a permanent **MAC address**, and Windows always runs the TCP/IP software, including the **loopback** address 127.0.0.1 (IPv4) and ::1 (IPv6) that the computer uses to talk to itself. Windows 11 lets you inspect these settings in four places: the **Settings** app, the classic **Control Panel**, **Command Prompt (CMD)**, and **PowerShell**. In this activity, you will explore all four on your own unit, observe how Windows behaves when no network is available, and then analyze a sample configuration taken from a computer that *is* connected to a network.

IV. Procedure

Reminder: This is a *viewing* activity. Do not change, disable, or reset any network setting. Close every dialog with **Cancel**. Record values exactly as Windows shows them.

Part A – Windows 11 Settings

1. Press **Windows + I** to open Settings, then click **Network & internet**. Write down the connection status shown at the top of the page (for example, "Not connected").
2. Click **Advanced network settings**. Under **Network adapters**, note every adapter listed (for example, Ethernet, Wi-Fi, Bluetooth Network Connection) and whether each one is enabled

or disconnected.

3. Scroll down and click **Hardware and connection properties**. For each adapter, record its **Name**, **Description**, **Physical address (MAC)**, and **Status** in **Table 1**.

Part B – Control Panel (Network Connections)

4. Press **Windows + R**, type **ncpa.cpl**, and press **Enter**. This opens the classic **Network Connections** window. A red X on an icon means the adapter has no network connection.
5. Right-click your **Ethernet** adapter and choose **Properties**. List the items checked under "This connection uses the following items" in **Table 2**.
6. Select **Internet Protocol Version 4 (TCP/IPv4)** and click **Properties**. Record whether the computer is set to **Obtain an IP address automatically** (DHCP) or **Use the following IP address** (static), and the same for DNS. Click **Cancel**, then **Cancel** again. Do not change anything.

Part C – Command Prompt (CMD)

7. Press the **Windows key**, type **cmd**, and press **Enter**. Run each command below, one at a time, and record the results in **Table 3**.

   hostname

   getmac /v

   ipconfig /all

   Tip: For a disconnected adapter, **ipconfig /all** shows "Media disconnected" and no IPv4 address, but it still shows the Description and Physical Address.

8. Test the computer's own TCP/IP software using the loopback address. Record whether you got replies and the time. For **ping localhost**, note the address that replied.

   ping 127.0.0.1

   ping localhost

9. Now observe what happens when there is no network. Record the **exact** message each command displays.

   ping 8.8.8.8

   nslookup google.com

10. View the ports your computer is listening on and its routing table. In Table 3, record three LISTENING local addresses with their port numbers, and the Netmask shown for the Network Destination 127.0.0.0.

    netstat -an | findstr LISTENING

    route print -4

11. Run the command below. It may take a few seconds. Scroll to **Network Card(s)** and record how many network cards are installed and the status of each.

    systeminfo

Part D – PowerShell

12. Right-click the **Start** button and choose **Terminal** (or press the **Windows key**, type **PowerShell**, and press **Enter**). Run each command below and record the results in **Table 4**.

    Get-NetAdapter

    Get-NetIPAddress -AddressFamily IPv4 | Format-Table InterfaceAlias, IPAddress, PrefixLength

    Test-Connection 127.0.0.1 -Count 4

    Get-NetTCPConnection -State Listen | Select-Object -First 5 LocalAddress, LocalPort

Tip: PowerShell commands follow a Verb-Noun pattern (Get-NetAdapter = "get the network adapters"). Compare each result with the matching CMD command from Part C.

Part E – Analyze a Networked Computer

13. The output below was taken from a laboratory computer that **is** connected to a network. Study it, then complete **Table 5**.

Sample Output 1. ipconfig /all (excerpt) from a networked PC

```
Host Name . . . :                 CICT-LAB-PC07
Ethernet adapter Ethernet:
Description . . . :               Realtek PCIe GbE Family Controller
Physical Address . . . :          3C-52-82-A1-7F-09
DHCP Enabled . . . :              Yes
Link-local IPv6 Address . . . :   fe80::8c2d:4a1f:93b2:61e5%12(Preferred)
IPv4 Address . . . :              192.168.10.37(Preferred)
Subnet Mask . . . :               255.255.255.0
Lease Obtained . . . :            Wednesday, October 14, 2026 8:02:15 AM
Lease Expires . . . :             Thursday, October 15, 2026 8:02:15 AM
Default Gateway . . . :           192.168.10.1
DHCP Server . . . :               192.168.10.1
DNS Servers . . . :               192.168.10.1, 8.8.8.8
```

Part F – Design a Network for Our Laboratory

14. Our laboratory does not yet have a network. In the box after Table 5, draw a simple **star topology** network for it. Include at least **four PCs**, one **switch**, one **router** (default gateway), one **printer**, and the **ISP / Internet**. Use lines to show connections.

15. Assign each device a private IPv4 address from the network **192.168.10.0/24**, using **192.168.10.1** for the router. Label each device in your drawing and complete the **IP Address Plan** (Table 6).

Part G – Optional / Bonus

16. In CMD, run **type C:\Windows\System32\drivers\etc\hosts** to *view* (not edit) the hosts file. In one or two sentences, explain what this file is for and how it relates to DNS.

V. Data and Findings

Table 1. Network Adapters in Settings (Part A)

Connection status shown on the Network & internet page: Connected

Adapter Name	Description	Physical Address (MAC)	Status
Local Area Connection* 1	Microsoft Wi-Fi Direct Virtual Adapter	c2:35:32:d0:5d:07	Not operational
Local Area Connection* 2	Microsoft Wi-Fi Direct Virtual Adapter #2	c2:35:32:d0:4d:17	Not operational
Wi-Fi	MediaTek Wi-Fi 6 MT7921 Wireless LAN Card	c0:35:32:d0:7d:27	Operational
Ethernet	Realtek PCIe GbE Family Controller	74:d4:dd:bb:c1:52	Not operational

Table 2. Adapter Properties (Part B)

Items Checked Under "This connection uses the following items"	**Item Checked (continued)**
Client for Microsoft Networks	Nested Network Virtualization
File and Printer Sharing for Microsoft Networks	Internet Protocol Version 4 (TCP/IPv4)
QoS Packet Scheduler	Microsoft LLDP Protocol Driver
Bridge Driver	Internet Protocol Version 6 (TCP/IPv6)

TCP/IPv4 Setting	**Automatic or Static?**
IP address	Automatic
DNS server address	Automatic

Table 3. Command Prompt Results (Part C)

Command / Item	**Your Findings**	**OSI Layer**
hostname	LAPTOP-ANEDJ6SL	Layer 7
getmac /v: MAC address and Transport Name	C0-35-32-D0-7D-27, \Device\Tcpip_{C5006CD3-809D-4AF3-A2C6-6393CEDE3EAC}	Layer 2
ipconfig /all: Media State of Ethernet	Media disconnected	—
ipconfig /all: DHCP Enabled	Yes	—
ping 127.0.0.1: replies and time	Reply from 127.0.0.1: bytes=32 time<1ms TTL=128	Layer 3
ping localhost: address that replied	::1	—
ping 8.8.8.8: exact message	Reply from 8.8.8.8: bytes=32 time=38ms TTL=118	Layer 3
nslookup google.com: exact message	Non-authoritative answer: Name: google.com Addresses: 2404:6800:4017:80a::200e 142.250.207.46	Layer 7
netstat -an: three LISTENING addresses:ports	0.0.0.0:135 0.0.0.0:445 0.0.0.0:4343	Layer 4
route print -4: Netmask for 127.0.0.0	255.0.0.0	—
systeminfo: number of NICs and status	2 NIC(s) Installed Media disconnected	—

Table 4. PowerShell Results (Part D)

Command	**What It Showed**	**Matches CMD? (Y/N)**
Get-NetAdapter: Name, Status, MacAddress, LinkSpeed	Wi-Fi, Up, C0-35-32-D0-7D-27, 130 Mbps	N
Get-NetIPAddress: IPv4 addresses and PrefixLength	Wi-Fi 192.168.1.102 24 Loopback Pseudo-Interface 1 127.0.0.1 8	N
Test-Connection 127.0.0.1: result	LAPTOP-ANE... 127.0.0.1 192.168.1.102 fe80::bc4:2b8b:406b:877%13 32 0	N
Get-NetTCPConnection -State Listen: two LocalAddress:LocalPort pairs	:: 58995	N

Table 5. Sample Output Analysis (Part E)

Question	**Your Answer**
IPv4 address of PC07	192.168.10.37
Subnet mask (and its /prefix)	255.255.255.0 (/24)
Default gateway, and what device it usually is	192.168.10.1
DNS servers	192.168.10.1, 8.8.8.8
Was the address assigned by DHCP or set manually? How do you know?	Assigned by DHCP. Because the output states **DHCP** Enabled . . . : Yes and provide lease acquisition dates
Public or private? Which RFC 1918 range?	Private. It belongs to the 192.168.0.0 - 192.168.255.255 range
IPv4 address of PC07 in binary (four octets)	11000000.10101000.00001010.00100101
MAC address, and the OSI layer where it is used	3C-52-82-A1-7F-09 used Layer 2
How long is the DHCP lease?	24 hours
If the DHCP server failed, what kind of address would PC07 assign itself?	APIPA

Figure 1. Proposed Laboratory Network Diagram (Part F)

Draw or paste your diagram here.

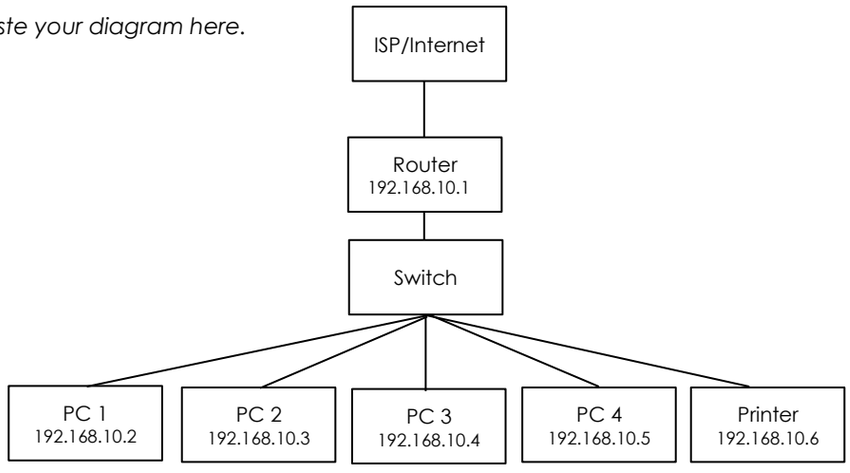


Table 6. IP Address Plan (Part F)

Device	IPv4 Address	Subnet Mask	Default Gateway
Router (gateway)	192.168.10.1	255.255.255.0	—
PC 1	192.168.10.2	255.255.255.0	192.168.10.1
PC 2	192.168.10.3	255.255.255.0	192.168.10.1
PC 3	192.168.10.4	255.255.255.0	192.168.10.1
PC 4	192.168.10.5	255.255.255.0	192.168.10.1
Printer	192.168.10.6	255.255.255.0	192.168.10.1

Part G Notes (Optional / Bonus)

In what I gathered the host files is more of a private address book that is stored on a computer. It maps directly to the numerical IP address. It relates to DNS because it acts like a local shortcut. The computer checks this file before asking an outside DNS server on the internet.

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Even without a network connection, what network information did your computer still have? Why do you think it still has it?

While having no network connection, my computer still has Wi-Fi profiles, network security passwords, hardware MAC addresses, and IP configurations.
The operating system saves these credentials so when it has access to a network or the Wi-Fi back turns on, it can authenticate and reconnect automatically without requiring to type the password.

2. Compare Settings, Control Panel, CMD, and PowerShell. Which tool did you find most useful for viewing network information, and why?

Each of these has its purpose, although, I think CMD is more useful in my experience, for the reason that it is for quickly viewing offline or online network information.

3. What did the messages from ping 8.8.8.8 and nslookup google.com tell you? How would you use messages like these when troubleshooting a real network?

The outputs from ping 8.8.8.8 and nslookup google.com provide two entirely different pieces of diagnostic information. ping 8.8.8.8 tests fundamental IP routing and physical connectivity, while nslookup google.com tests the Domain Name System (DNS) resolution.

4. Why should you be careful about sharing your MAC address, IP address, or ipconfig /all screenshots online? Relate your answer to data privacy.

Sharing screenshots of ipconfig /all readout or raw network details online poses a risk to the data privacy. A single IP or MAC address might seem like harmless technical gibberish, to a malicious actor or data broker, this information acts as pieces of a puzzle used to identify, locate, and track someone.

VII. Scoring Guide

Criteria	Description	Points
Completeness	All tables (1–6) and required fields are filled out.	25
Accuracy	Recorded values match what the tools show and are consistent across Settings, CMD, and PowerShell; sample analysis, binary conversion, and OSI layers are correct.	30
Network Design	Diagram shows a star topology with PCs, switch, router, printer, and ISP; IP plan uses valid, non-duplicate addresses in 192.168.10.0/24.	15
Realization / Reflection	Answers are thoughtful, based on the student's own results, and well written.	20
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	10
Total		**100**